Institutional Ownership Networks and Return Co-movement During Market Stress

Varshini Narayanan

Research Question

Do mega-cap stocks with similar institutional ownership structures exhibit stronger return co-movement during periods of market stress?

Underlying Mechanism:

During volatility shocks, institutional investors may reduce equity exposure or rebalance portfolios toward perceived safer assets

If the same institutions hold multiple firms, these adjustments can generate common trading flows, causing those firms' prices to move together.

Hypothesis:

Firms that share more institutional investors or are more central in the institutional ownership network will exhibit higher return correlations during market stress.

Description of the Topic

- My project examines whether mega-cap firms with similar institutional investor ownership structures move together during periods of market stress.
- Using institutional holdings data from 13F filings, I constructed an ownership network linking firms that share institutional investors.
- I then tested whether firms with greater institutional ownership overlap exhibit stronger return correlations during high-volatility market periods.

Data

Data Source	Sample Size	Time Period	Collection Method	Key Variables Created
Yahoo Finance Screener	List of largest U.S. mega-cap firms	Current market snapshot	Web scraping (BeautifulSoup)	Firm tickers
WRDS – CRSP Daily Stock Data	Daily observations for selected mega-cap firms	Study period used in analysis	WRDS SQL query	Daily firm returns, daily market returns, firm identifiers (PERMNO/CUSIP), quarter variable created from date
WRDS – 13F Institutional Holdings (Thomson/Refinitiv S34)	Institutional holdings across institutions and firms	Same study period	WRDS SQL query	Institutional ownership %, shares held, institutional identifier (mgrno), ownership by quarter
FRED API – S&P 500 Data	Daily S&P 500 observations	Same study period	FRED API	Market return, market_realized_vol

Data Cleaning Takeaway

Challenge:

Institutional ownership (13F) is reported quarterly, while stock returns and volatility measures are daily.

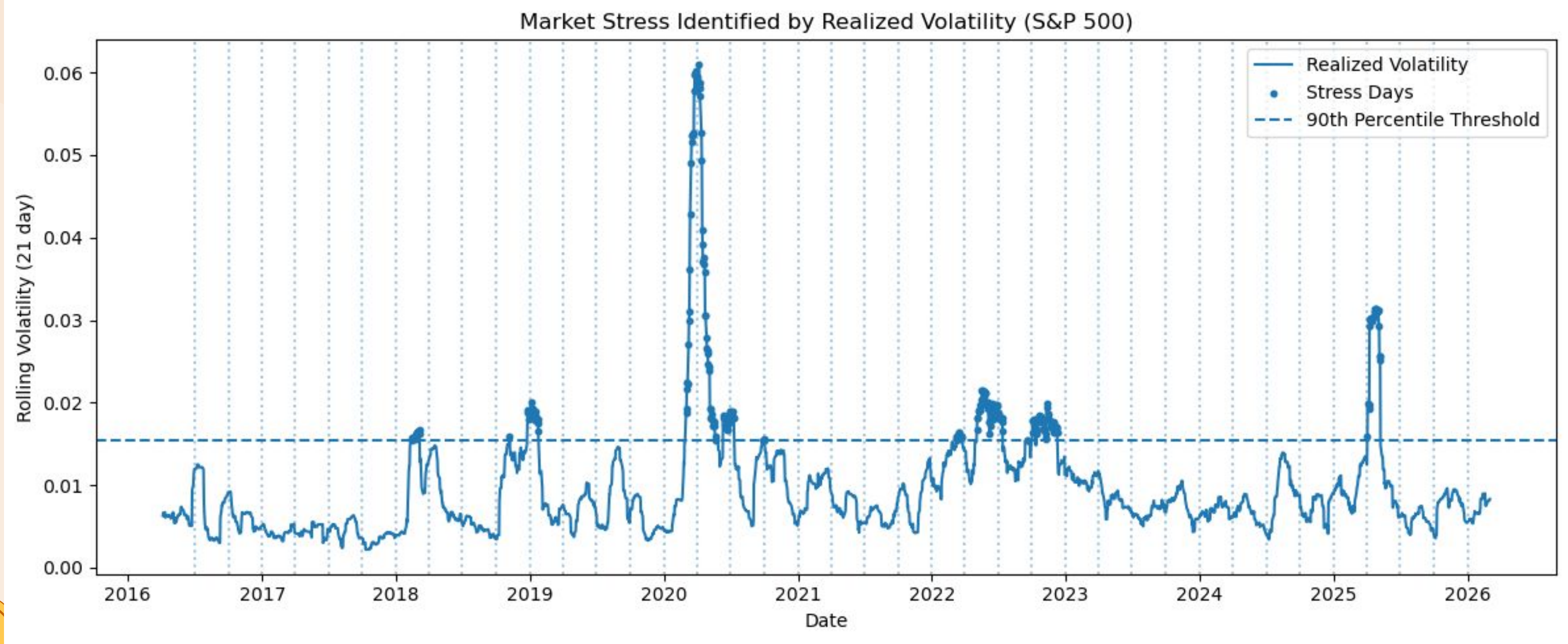
I initially created quarter start and end date variables to align the datasets.

This added unnecessary complexity and led to mixing up quarter boundaries with volatility dates, producing merged tables with missing values.

Solution:

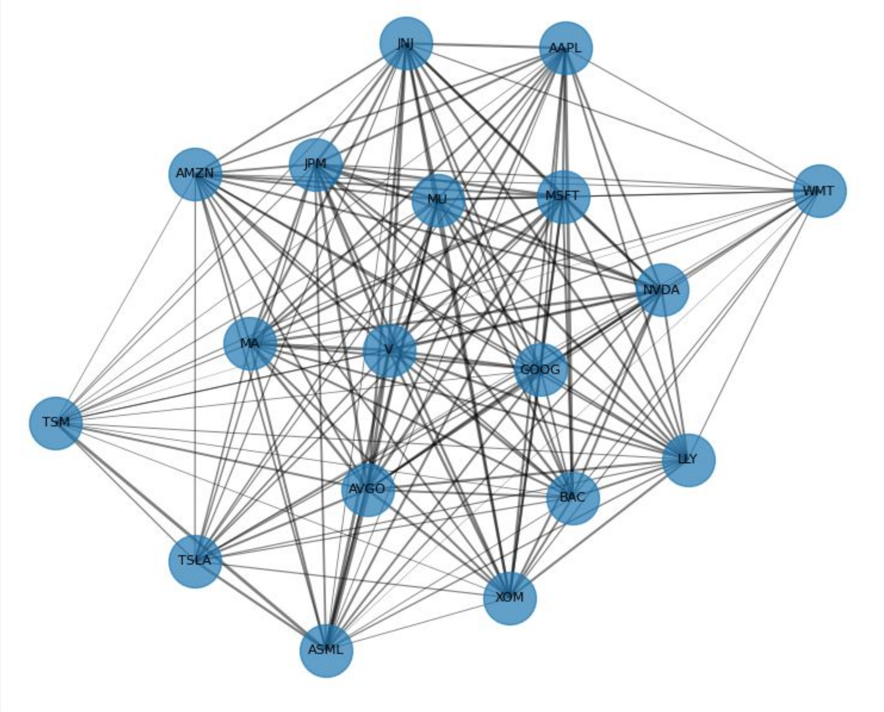
I simplified the merge by using the 13F filing/reporting date instead of storing multiple quarter boundary variables.

Stress Days

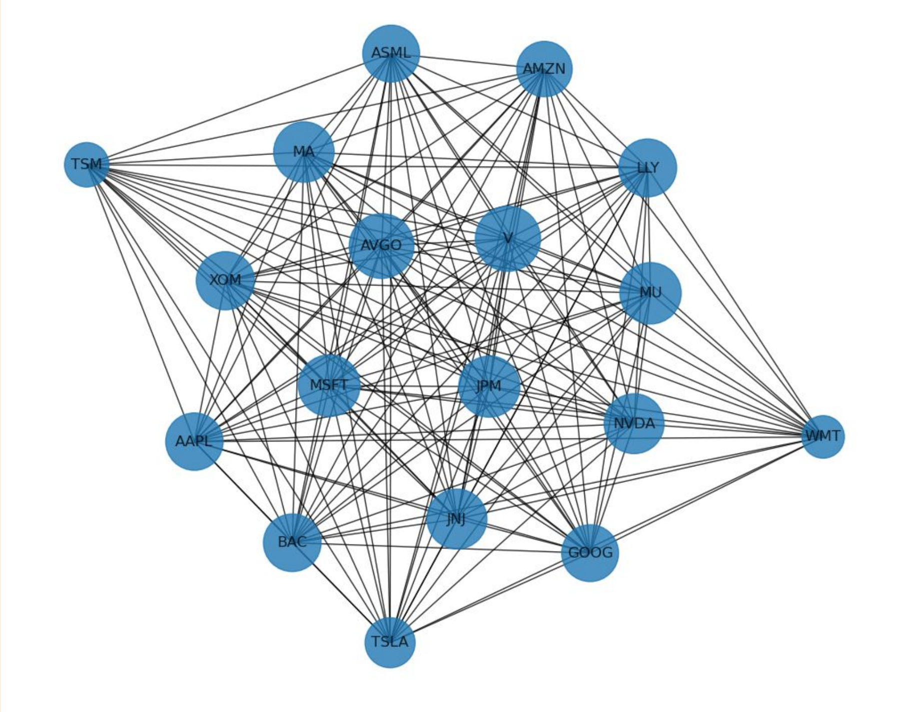

Market Stress Identified by Realized Volatility (S&P 500)
Realized Volatility
Stress Days
90th Percentile Threshold
Rolling Volatility (21 day)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Date

Ownership Networks – 2020Q1

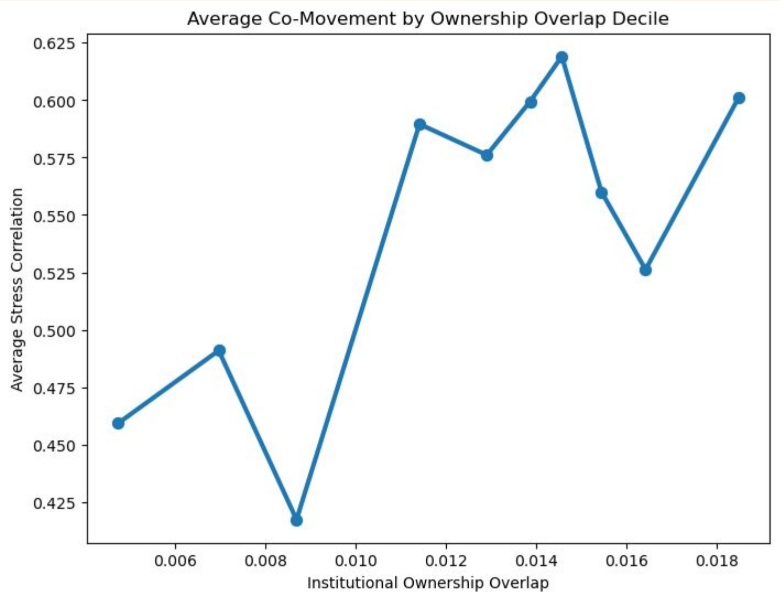
Edge thickness = ownership overlap



Node size = centrality



Ownership Overlap and Co-Movement



Corr_ij = β0 + β1 * OwnershipOverlap_ij + β2 * SameSector_ij + ε_ij

```
                            OLS Regression Results
==============================================================================
Dep. Variable:      stress_correlation   R-squared:                       0.098
Model:                             OLS   Adj. R-squared:                  0.097
Method:                  Least Squares   F-statistic:                     294.7
Date:                 Wed, 04 Mar 2026   Prob (F-statistic):          5.40e-118
Time:                         09:19:41   Log-Likelihood:                -764.91
No. Observations:                 3134   AIC:                             1536.
Df Residuals:                     3131   BIC:                             1554.
Df Model:                            2
Covariance Type:                   HC3
=====================================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                 0.3721      0.018     20.492      0.000       0.337       0.408
ownership_overlap    10.5654      1.347      7.845      0.000       7.926      13.205
same_sector           0.2257      0.010     21.906      0.000       0.206       0.246
=====================================================================================
Omnibus:                       667.196   Durbin-Watson:                   1.091
Prob(Omnibus):                   0.000   Jarque-Bera (JB):             1477.556
Skew:                           -1.209   Prob(JB):                         0.00
Kurtosis:                        5.339   Cond. No.                         241.
==============================================================================
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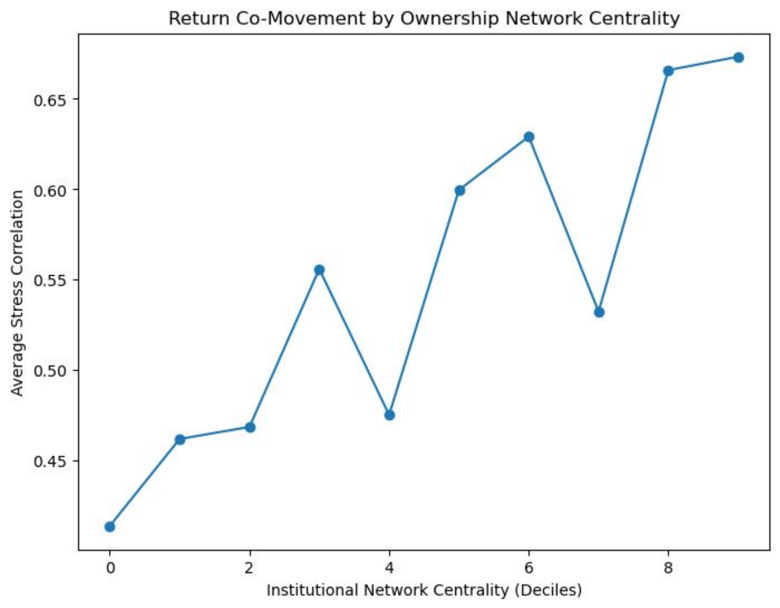
Network Centrality Regression



Corr_ij = β0 + β1 * PairCentrality_ij + β2 * SameSector_ij + ε_ij

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     stress_correlation   R-squared:                       0.123
Model:                            OLS   Adj. R-squared:                  0.122
Method:                 Least Squares   F-statistic:                     309.1
Date:                Wed, 04 Mar 2026   Prob (F-statistic):          6.98e-123
Time:                        09:00:32   Log-Likelihood:                -705.08
No. Observations:                3026   AIC:                             1416.
Df Residuals:                    3023   BIC:                             1434.
Df Model:                           2
Covariance Type:                  HC3
===================================================================================
                      coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------
const              -0.0142      0.046     -0.310      0.756      -0.104       0.075
pair_centrality     2.4121      0.207     11.626      0.000       2.005       2.819
same_sector         0.2080      0.011     19.563      0.000       0.187       0.229
==============================================================================
Omnibus:                      679.085   Durbin-Watson:                   1.065
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1612.105
Skew:                          -1.239   Prob(JB):                         0.00
Kurtosis:                       5.578   Cond. No.                         37.8
==============================================================================
```

Conclusion

Answer to Research Question:

Firms with greater shared institutional ownership exhibit stronger return co-movement during market stress.

Firms that are more central in the institutional ownership network also show higher correlations with other firms.

Conclusion:

Institutional ownership networks may amplify co-movement across firms during market stress.

Next Step:

Compare these relationships during non-stress periods to determine whether the effect is stronger specifically during market stress.